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Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

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1. Introduction

The formation water volume factor, B_w , represents the volume of water at reservoir conditions required to produce one barrel of water at standard surface conditions. In other words, it relates the volume occupied by formation water in the reservoir to its corresponding volume at the surface:

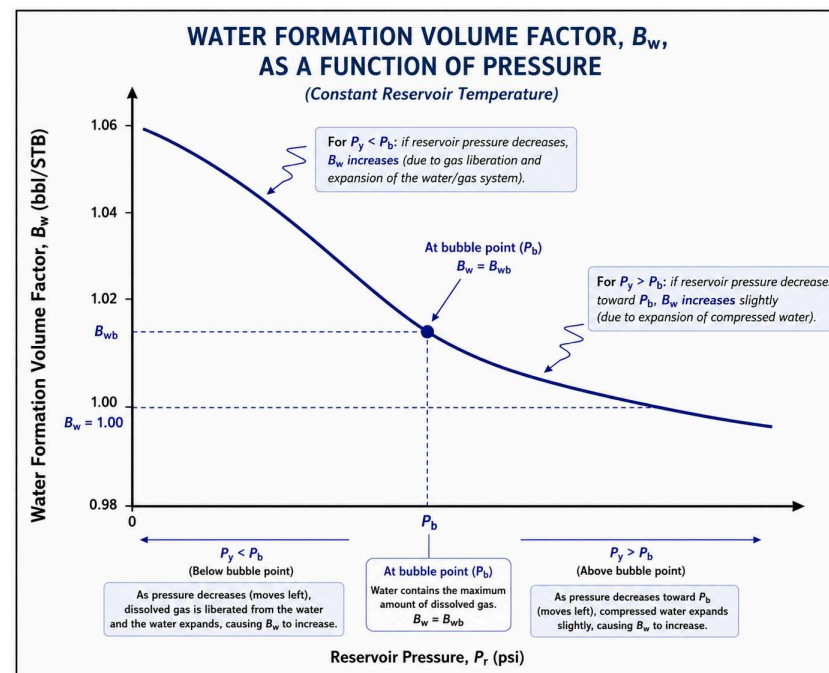
$$B_w = \frac{\text{volume of water at reservoir conditions}}{\text{volume of water at stock-tank conditions}}$$

Similar to the oil formation volume factor, B_o , the behavior of the formation water volume factor, is influenced by three main effects:

1. Release of dissolved gas: As pressure decreases, a small amount of gas dissolved in the formation water may be released.
2. Expansion of formation water: As pressure decreases, formation water expands slightly because of its finite compressibility.
3. Thermal contraction: As temperature decreases from reservoir to surface conditions, the formation water contracts, reducing its volume.

The solubility of natural gas in formation water is substantially lower than its solubility in crude oil. Consequently, changes in dissolved-gas content generally have only a minor effect on B_w , particularly when compared with the influence of dissolved gas on the oil formation volume factor B_o . Similarly, water is only slightly compressible. Therefore, the volumetric changes caused by pressure variations are relatively small. Temperature changes can also affect the water volume through thermal expansion or contraction, but under typical reservoir conditions these effects remain moderate. As a result, the formation water volume factor usually remains relatively close to unity and varies much less than (B_o). Typical values are approximately 1.06 RB/STB, although the actual value depends on pressure, temperature, salinity, and dissolved-gas content.

The following figure illustrates the typical behavior of B_w as a function of pressure at constant reservoir temperature. Because formation water is only slightly compressible, B_w changes gradually with pressure rather than exhibiting the large volumetric variations commonly observed for reservoir oils.



There are several numerical correlations available to compute the oil formation volume factor, B_o , some of the most widely used include:

- McCain
- McCoy

2. Numerical solution

$p < p_{wb}$

$B_w = (1 + \Delta V_{wp}) (1 + \Delta V_{wT})$

$$\Delta V_{wT} = -1.001 \times 10^{-2} + 1.33391 \times 10^{-4} T + 5.50654 \times 10^{-7} T^2$$

$\Delta V_{wp} = -1.95301 \times 10^{-9} p T - 1.72834 \times 10^{-13} p^2 T - 3.58922 \times 10^{-7} p - 2.25341 \times 10^{-10} p^2$

$p > p_{wb}$

$B_w = B_{wb} \times \exp[c_w(p_{wb} - p)]$

2.1 Import libraries

```
In [2]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [3]: gamma_g=0.9
gamma_o=31
T=np.array([100,110,120])

pb=2200
Bob=1.3
```

```
cw=5*10**-6
p=np.linspace(100,3000,100)
```

2.3 Calculation and plotting

Note: To calculate the oil formation volume factor B_o is required to know the solution gas-oil ratio R_s . Please review the corresponding notebook

```
In [6]: plt.figure(dpi=300, figsize=(8,4))

for i in range(len(T)):

    # Properties at bubble point
    deltatb=-1.001*10**-2+1.3391*10**-4*T[i]+5.50654*10**-7*T[i]**2
    deltapb=-1.95301*10**-9*pb*T[i]-1.72834*10**-13*pb**2*T[i]-3.58922*10**-7*pb-2.25341*10**-10*pb**2

    Bwb = (1+deltapb)*(1+deltatb)

    Bw_values = []

    for j in range(len(p)):

        # Bo at pressure Lower than pb
        if p[j] < pb:
            deltat=-1.001*10**-2+1.3391*10**-4*T[i]+5.50654*10**-7*T[i]**2
            deltap=-1.95301*10**-9*p[j]*T[i]-1.72834*10**-13*p[j]**2*T[i]-3.58922*10**-7*p[j]-2.25341*10**-10*p[j]**2

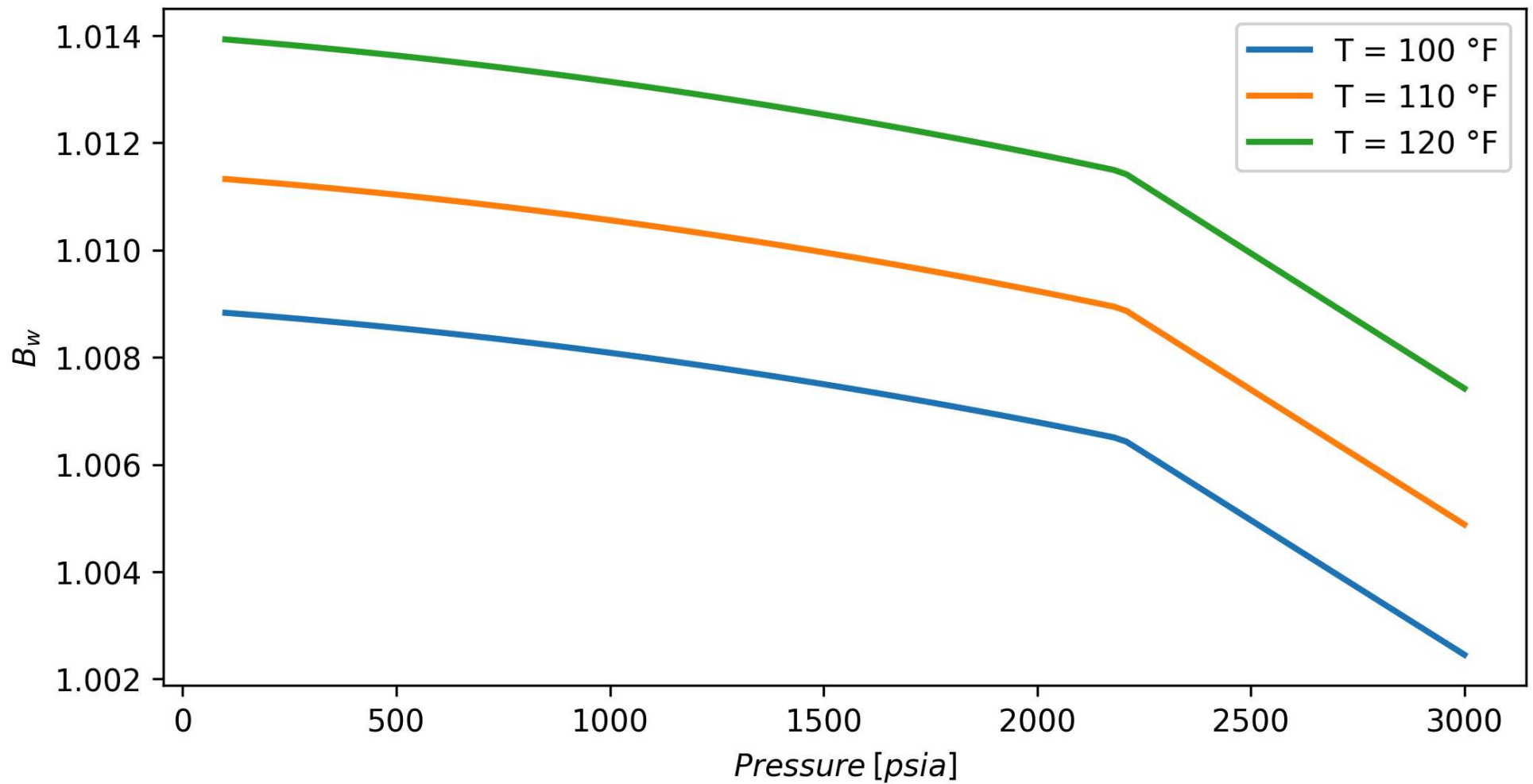
            Bw = (1+deltap)*(1+deltat)

        # Bo at pressure greater than pb
        else:
            Bw = Bwb*np.exp(cw*(pb - p[j]))

        Bw_values.append(Bw)

    plt.plot(p, Bw_values, linestyle='-', linewidth=2,
             label=f'T = {T[i]} °F')

plt.xlabel('$Pressure\:[psia]$')
plt.ylabel('$B_w$')
# plt.grid(True)
plt.legend()
plt.show()
```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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In []: